

Mapping Ecological Sites with Deep Learning

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Introduction

Across the western United States, land managers, ranchers, and conservation agencies rely on a system of land classification called Ecological Sites. An Ecological Site is a classification of land: a patch of landscape with a distinct combination of soil, climate, and topography that determines what plants can grow there and how the land responds to disturbances. It describes the whole soil-climate-vegetation system and how it can shift between different states over time (Karl and Herrick, 2010). Ecological site maps are built largely by hand; soil scientists and ecologists visit sites, take measurements, and use their expert judgment to draw boundaries and assign classifications. This is accurate but slow and expensive, and it means large parts of the country only have coarse, low-resolution versions of these maps (Padarian et al., 2019; Nauman et al., 2026). Our project asks whether a deep learning model can help predict these classifications from data that's already publicly available (satellite imagery, elevation data, climate records, and existing soil surveys), and if it can classify at a finer resolution

Research question:

Within a Major Land Resource Area (MLRA) (a USDA-defined zone with broadly similar climate and geology), can a model that looks at spatial patterns in the landscape, using remote sensing and publicly available data, predict which Ecological Site a given location belongs to.

A follow-up question: does a model that considers the surrounding landscape (not just a single point) do meaningfully better than a simpler model that only looks at point-level values?

The Problem:

Ecological Site maps are foundational to land management decisions, but they are incomplete or outdated across much of the country because manual mapping is costly and time consuming. If we can show that a model can reasonably predict these classifications from data that's already collected (satellite imagery, elevation models, NRCS soil data), this will allow for finer resolution mapping without the costs of manually collecting data. This is the first step towards asking whether the same approach could eventually work at a larger scale (the entire continental US) to understand what environmental factors (climate, terrain, soil, or vegetation) matter most in determining ecological potential. We're deliberately starting small (one MLRA) to test the method before trying to scale up.

Methods:

1. We will use MLRA 042B as our study area. Ecological Site classifications for this MLRA are published by the USDA's Natural Resources Conservation Service (NRCS). We'll

collect four types of publicly available data, all as map layers (rasters images .tif) covering the same area:

1. Climate data (temperature, precipitation) from PRISM, a widely-used US climate dataset.
2. Terrain data (slope, aspect, how water flows across the land) calculated from elevation models from <https://www.usgs.gov/the-national-map-data-delivery/gis-data-download>.
3. Soil data from the USDA's SSURGO soil survey database.
4. Satellite imagery (from USGS Earth Explorer, Landsat or Sentinel satellites), which shows vegetation greenness and how it changes across seasons.
5. Open to other data input ideas

2. Models:

XGBoost: A non-deep-learning technique that builds many small decision trees one after another, where each new tree tries to correct the mistakes of the previous ones. It works on each location's covariate values point-by-point, without looking at the surrounding landscape. It's widely used in digital soil and land mapping (Chen & Guestrin, 2016; Meles et al., 2026).

Convolutional Neural Network (CNN): A CNN is a type of neural network that looks at small square "patches" of the map around each location for example, an 11x11 pixel neighborhood, instead of just a single point. This lets it pick up on spatial patterns, like how a hillside's shape or a valley's shading might influence the ecological classification, that a point-based model would miss entirely (Padarian et al., 2019).

3. Because nearby locations tend to be similar to each other (spatial autocorrelation), we can't just randomly split our data into training and testing sets, doing so would let the model "cheat" by learning from a location's near-twin. Instead, we'll use a method called spatial cross-validation, which makes sure the test locations are geographically separated from the training locations, giving us an honest measure of how well the model generalizes to a new area (Roberts et al. 2017).
4. **BONUS:** Beyond just checking which model is more accurate, it would be interesting to use a technique called SHAP (SHapley Additive exPlanations), which measures how much each input (climate, terrain, soil, imagery) contributed to each individual prediction. This lets us answer a more interesting question than "which model wins": which environmental factors drive the classification in this region (Wadoux et al., 2020)?

We're only studying one MLRA, our results will mostly reflect how soil, terrain, and vegetation vary within a consistent climate. We'll treat this project explicitly as a methods test, not a final answer. We'll train the models to predict the correct Ecological Site classification for locations across our study region, using existing NRCS records as the "ground truth" labels to train the model.

References

- Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. KDD.
<https://arxiv.org/abs/1603.02754>
- Karl, J. W., & Herrick, J. E. (2010). Monitoring and assessment based on ecological sites. *Rangelands*, 32(6), 60–64. <https://doi.org/10.2111/RANGELANDS-D-10-00082.1>
- Meles, M. B., Guertin, D. P., Burns, I. S., Goodrich, D. C., Hernandez, M., Metz, L. J., Saeedimoghaddam, M., Ponce-Campos, G., Nearing, G., Williams, J., Barker, S., Houdeshell, C.-A., & Archer, S. R. (2026). From expert knowledge to data-driven landscape classification: Mapping ecological site groups across climatic and edaphic gradients. *Landscape Ecology*, 41, Article 105.
<https://doi.org/10.1007/s10980-026-02385-0>
- Nauman, T. W., Brungard, C. W., Salley, S. W., & Allred, B. W. (2026). Image-based deep learning enables flexible, holistic 3D mapping of multiple soil properties. *Geoderma*, 473, 117985.
<https://sciencedirect.com/science/article/pii/S0016706126003137>
- Padarian, J., Minasny, B., & McBratney, A. B. (2019). Using deep learning for digital soil mapping. *SOIL*, 5(1), 79–89. <https://soil.copernicus.org/articles/5/79/2019/>
- Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Aroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., & Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8), 913–929. <https://doi.org/10.1111/ecog.02881>
- Wadoux, A. M. J.-C., et al. (2020). Game theory interpretation of digital soil mapping convolutional neural networks. *SOIL*, 6(2), 389–397.